

axi_lite_read_channel_decoder.v

AUTHORS

JAY CONVERTINO

DATES

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INFORMATION

Brief

Verify read address and contain holdbuffer interface. Valid address? Allow output till invalid address presented.

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axi_lite_read_address_decoder

AXI lite address recoder for read channel.

Parameters

ADDRESS_WIDTH	Width of the AXI LITE address port in bits.
BUS_WIDTH	Width of the AXI LITE bus data port in bytes.
DATA_BUFFER	Buffer data channel, 0 to disable.
TIMEOUT_BEATS	Number of clock cycles (beats) to count till timeout. 0 disables timeout.
SLAVE_ADDRESS	Array of Addresses for each slave (0 = slave 0 and so on).
SLAVE_REGION	Region for the address that is valid for the SLAVE ADDRESS.

Ports

connected	Core has established channel connection
aclk	Input clock
arstn	Input negative reset
s_axi_araddr	Slave read chanel input address.
s_axi_arprot	Slave read chanel input address protection
s_axi_arvalid	Slave read chanel input address is valid
s_axi_arready	Slave read chanel input is ready.
s_axi_rdata	Slave read chanel input data.
s_axi_rresp	Slave read chanel input data response.
s_axi_rvalid	Slave read chanel input data valid
s_axi_rready	Slave read chanel input is ready.
m_axi_araddr	Master read chanel output address.
m_axi_arprot	Master read chanel output address protection
m_axi_arvalid	Master read chanel output address is valid
m_axi_arready	Master read chanel output is ready.
m_axi_rdata	Master read chanel output data.
m_axi_rresp	Master read chanel output data response.
m_axi_rvalid	Master read chanel output data valid
m_axi_rready	Master read chanel output is ready.

INSTANTIATED MODULES

inst_addr_buffer

Buffer for the address

inst_addr_verify

Decoder for address bus.

inst_data_resp_buffer

If data buffer enabled, this holdbuffer will be generated.